

# Supplementary Material

## *Catalogue of Catalogues: A Web Platform for Earthquake Catalogue Management, Merging, and Statistical Analysis*

This supplement lists the parameters of the synthetic catalogues used in the worked example of the main paper (Table S1) and collects additional screen captures of the *Catalogue of Catalogues* (CofC) platform that support, but are not essential to, the argument of the main paper. All captures are from the release described in the main paper, running in a browser against a local instance.

**Data.** Every capture uses a reduced-scale version of the worked example’s synthetic pair, built by the same seeded generator (`paper/figures/synthetic_catalogues.py`, seed 42) with each true earthquake kept with probability 1/23: a 4,922-event “GeoNet-like” catalogue and a 4,411-event “Agency B” catalogue, each uploaded as a QuakeML 1.2 file written by `python paper/figures/generate_figures.py --screenshot-catalogues`. The values quoted in the captions are those of this reduced pair, computed by the platform’s own code (`paper/figures/screenshot_expectations.ts`, which parses the files with the upload parser and drives the Analysis page’s analysis worker); the worked example of the main paper uses the full-scale pair. No real event data are shown anywhere in this supplement.

## Synthetic catalogue construction

Table S1: Inputs of the synthetic worked example. Everything else the main paper reports about it (catalogue sizes, overlap, quality scores, duplicates,  $M_c$ ,  $b$ -values and declustering counts) is computed from the resulting catalogues by the platform.

| Component         | Specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Period and domain | 2020-01-01 to 2024-12-31 (1827 days); true epicentres in 166°–179°E, 47.5°–34.5°S.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Background        | $10^6$ independent events, uniform in time, magnitudes from a Gutenberg-Richter law with $b = 1.00$ (continuous, $M0.5$ –7.5), about 3200 events of $M \geq 2.3$ per year.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Source zones      | Share of the background, centre, and Gaussian spread along / across strike (strike azimuth): Hikurangi margin 19%, (177.3°E, 40.0°S), 170/45 km (35°), depth 8–30 km; central North Island crust 22%, (176.0°E, 39.0°S), 150/60 km (40°), 2–15 km; North Island slab 16%, (175.8°E, 39.6°S), 190/60 km (40°), 40 km plus an exponential tail of scale 30 km (at most 250 km); Marlborough–Canterbury 16%, (173.2°E, 42.4°S), 130/60 km (50°), 3–20 km; Alpine Fault 14%, (170.5°E, 43.6°S), 180/50 km (55°), 2–15 km; Fiordland–Puysegur 8%, (166.9°E, 45.6°S), 110/45 km (25°), 5 km plus an exponential tail of scale 25 km (at most 150 km); distributed offshore 5%, uniform over the domain, 5–30 km. |
| Aftershocks       | Sequences on every background event of $M \geq 4.5$ , sized so that 23% of the true events of $M \geq 2.0$ are aftershocks and shared between sequences in proportion to $10^{0.8(M-4.5)}$ (Utsu, 1970); times from the modified Omori law ( $p = 1.1$ , $c = 0.05$ d, up to 365 d; Omori, 1894; Utsu, 1961); epicentres uniform in a disc of radius twice the rupture length $\log_{10} L = 0.59M - 2.44$ (Wells and Coppersmith, 1994); depth of the mainshock $\pm 3$ km; magnitudes from a law with $b = 1.30$ truncated 1.2 units below the mainshock (Båth, 1965).                                                                                                                                   |
| Station networks  | Jittered grids on New Zealand land (Natural Earth 1:10m): GeoNet-like, $0.36^\circ \times 0.47^\circ$ ( $\sim 40$ km), 166 stations over the whole country; Agency B, $0.27^\circ \times 0.35^\circ$ ( $\sim 30$ km), 67 stations in 175.5°–178.6°E, 41.4°–37.6°S.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Recording         | A station records an earthquake if its hypocentral distance is below $R(M) = 10^{1.11+0.38M}$ km times a log-normal station factor ( $\sigma_{\ln} = 0.25$ ); an agency catalogues every earthquake that at least four of its stations record. Each recording station adds a P phase, and an S phase with probability 0.6.                                                                                                                                                                                                                                                                                                                                                                                 |

| Component         | Specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reported metadata | From the recording stations: used-station and phase counts, azimuthal gap (largest gap between their azimuths), nearest-station distance $\Delta_{\min}$ . Horizontal uncertainty $\sigma_h = 0.9\sqrt{6/n}e^{\max(0, \text{gap}-150^\circ)/70^\circ}(1 + \Delta_{\min}/80 \text{ km})\epsilon$ km; depth uncertainty $\sigma_z = 1.6\sigma_h(1 + \Delta_{\min}/(z + 5 \text{ km}))\epsilon$ , with the depth fixed (operator assigned: 12 km GeoNet-like, 10 km Agency B) and no depth uncertainty reported where $\sigma_z > 20$ km; origin-time uncertainty $0.05 + \epsilon\sigma_h/7$ s; RMS residual with median 0.20 s (GeoNet-like) or 0.14 s (Agency B), times $(1 + \Delta_{\min}/200 \text{ km})$ ; magnitude stations $n_{\text{amp}} \approx 0.6\text{--}0.95n$ and magnitude uncertainty $0.05 + 0.30/\sqrt{n_{\text{amp}}}$ . Each $\epsilon$ is an independent log-normal factor ( $\sigma_{\ln} = 0.25$ ; 0.2 for the time uncertainty), and the RMS residual varies log-normally about its median with $\sigma_{\ln} = 0.3$ . |
| Reported values   | True epicentre, depth, origin time and magnitude plus errors of exactly the reported size; $M_L$ , evaluation mode manual and status reviewed for both agencies; magnitudes to full precision (GeoNet-like) or to 0.1 (Agency B).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

# Statistical analysis in the browser

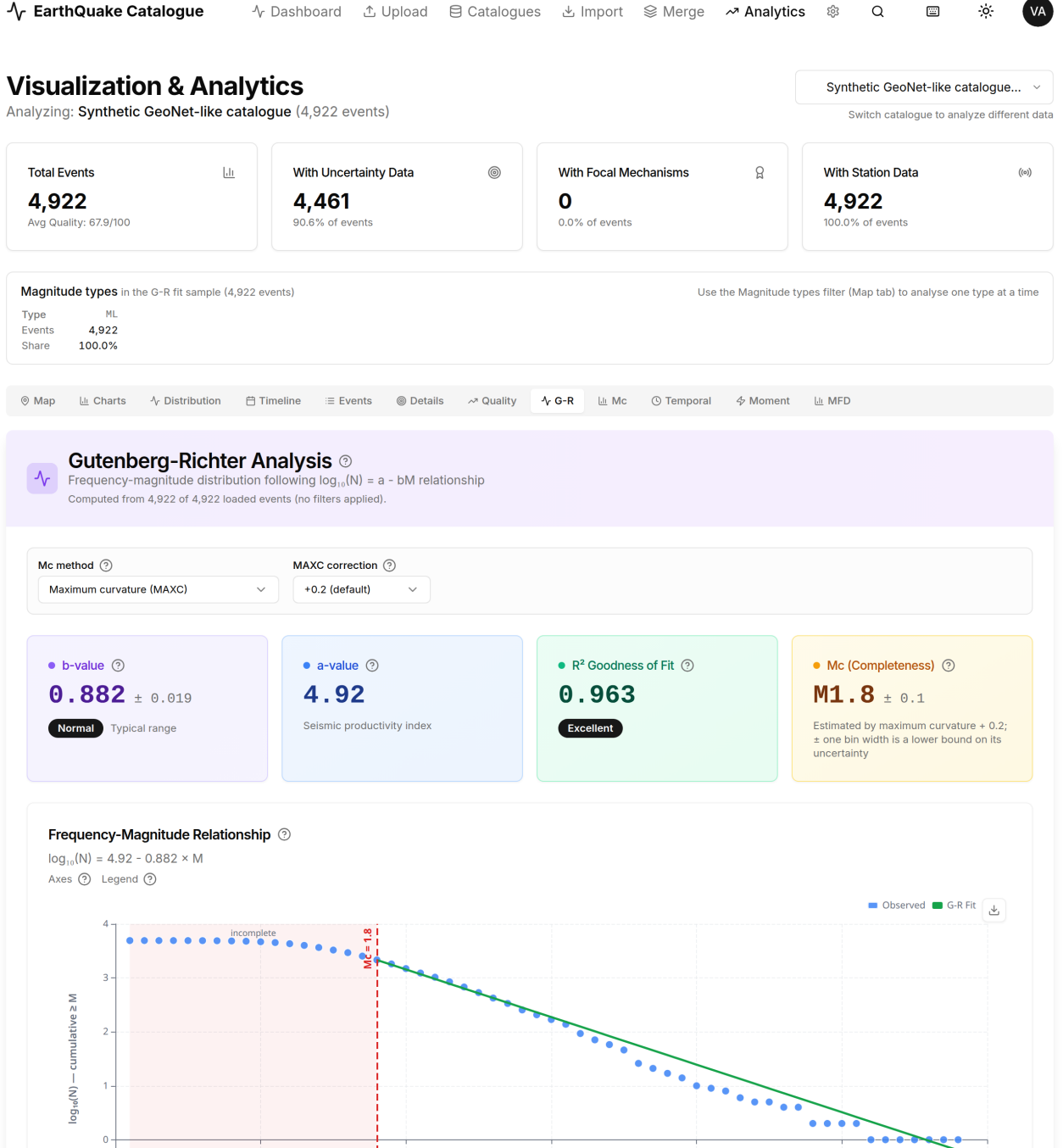


Figure S1: Gutenberg-Richter analysis of the reduced GeoNet-like catalogue, computed client-side with the default  $M_c$  settings (maximum curvature plus 0.2). The platform reports  $\hat{b}$  with its counting-statistics uncertainty, the  $a$ -value, the goodness of fit and  $M_c$ ; the frequency-magnitude plot marks the range below  $M_c$  and overlays the fitted relation. Above  $M_c = M1.8 \pm 0.1$  the fit gives  $\hat{b} = 0.882 \pm 0.019$  ( $N = 2,151$ ;  $a = 4.92$ ,  $R^2 = 0.963$ ). As in the worked example of the main paper, this lies below the  $b = 1.00$  planted in the synthetic background, because the synthetic networks are not complete at  $M1.8$  offshore and at depth: a single global  $M_c$  is set by the best-covered part of the catalogue. The magnitudes are reported to full precision, so no binning correction is applied. The capture is a check of the platform's behaviour on a controlled input, not a measurement of New Zealand seismicity.

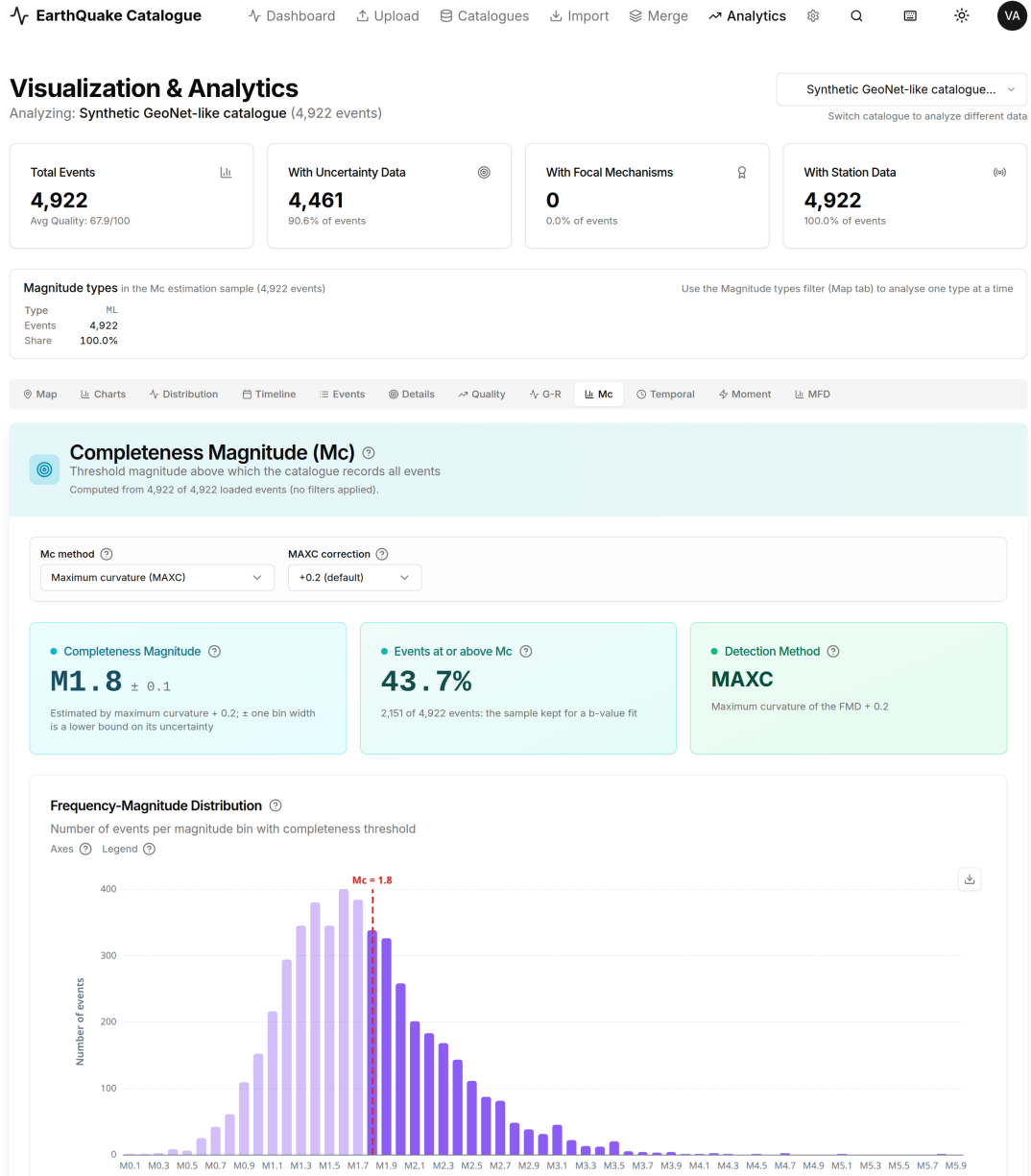


Figure S2: Magnitude-of-completeness panel for the same catalogue, with the default settings (maximum curvature, correction +0.2; the goodness-of-fit test is the alternative method, and the correction can be set from 0 to 0.5). The fullest bin of the non-cumulative distribution is  $M1.6$ , so  $M_c = M1.8 \pm 0.1$  (one bin width, a lower bound on its uncertainty), and 43.7% of the events (2,151 of 4,922) lie at or above it. That share is not a completeness score: for a perfectly complete catalogue with  $b = 1$  the +0.2 correction alone would limit it to about 63%. The estimate is shown with its method and correction, so that the completeness assumption behind any subsequent  $b$ -value is visible at the point of use.

# Event-level quality index

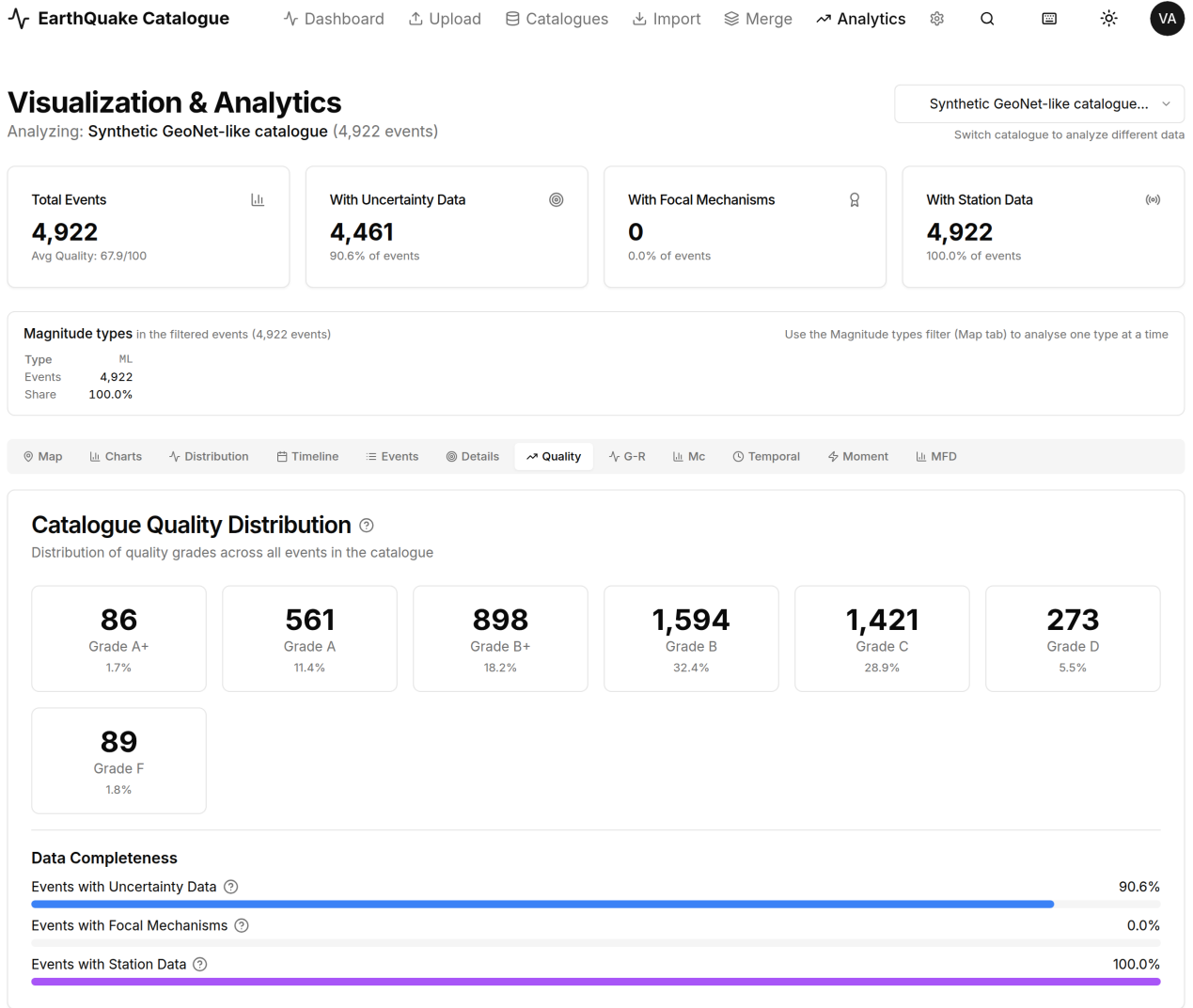


Figure S3: Catalogue-level quality summary for the same catalogue. The grade distribution and the mean score (67.9) are the five-dimension per-event index of equation 1 of the main paper, which scores location, network geometry, solution, magnitude, and evaluation status; every event is scored when it is stored and mapped to an A<sup>+</sup>–F grade. The synthetic reports carry the fields equation 1 reads (horizontal, depth and origin-time uncertainties, azimuthal gap, station and phase counts, RMS residual, magnitude uncertainty and station count, and an analyst-reviewed evaluation status), so the grades reflect the recording geometry: 86 events (1.7 %) reach A<sup>+</sup>, most events are graded B (32.4 %) or C (28.9 %), and the lowest grades go to offshore and small events recorded by few stations. The field-completeness bars are a separate catalogue-level metric: 90.6 % of the events have a location uncertainty in the sense of the “With Uncertainty Data” card (a latitude, longitude or depth uncertainty), the others having a fixed depth and only a horizontal uncertainty, and none has a focal mechanism, which equation 1 does not use.

## References

- Båth, M. (1965). Lateral inhomogeneities of the upper mantle, *Tectonophysics* **2**, no. 6, 483–514.
- Omori, F. (1894). On the after-shocks of earthquakes, *Journal of the College of Science, Imperial University of Tokyo* **7**, 111–200.
- Utsu, T. (1961). A statistical study on the occurrence of aftershocks, *Geophysical Magazine* **30**, 521–605.

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- Wells, D. L., and K. J. Coppersmith (1994). New empirical relationships among magnitude, rupture length, rupture width, rupture area, and surface displacement, *Bulletin of the Seismological Society of America* **84**, no. 4, 974–1002.